Edge acoustic & Vibration Condition Monitor
STM32 FHIICEUB

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Problem: Faults in small rotating machines change vibration and sound well before a breakdown. Listening by ear is subjective and industrial motors are costly. Our low-cost edge monitor learns healthy 6V/12V DC motor behaviour to automatically flag faults like imbalance, loose mounting, or load changes.

How it works: The STM32 FHIICEUG collects vibration data from the MPU6050 (IDC, 1kHz) and sound data from the INMP441 (I2S+DMA, 16kHz). Using CMSIS-DSP, each window is processed to calculate RMS, Peak-to-peak and dominant frequency. Twenty healthy windows establish the baseline (mean and standard deviation). During monitoring, if any feature has $|z| > 3$ for 3 consecutive windows, the system detects an ABNORMAL condition; otherwise it remains NORMAL.

Outputs: OLED displays stats and features; LED & buzzer indicate alerts. CSV data is sent via UART / Bluetooth. KEY button or Bluetooth commands control operation. TB6612FNG drives the motor using 10kHz PWM.

Concepts used: TIM3 TRGO, DMA, NVIC, Peripherals, Interrupts, main loop flags, ISR flags handling and sample storage, main-loop FFT processing and decision-making.

Fault signatures: Imbalance raises vibration RMS, loose mounting adds 2x/3x harmonics.

Next steps: Any ML classifier is planned for the extension.

System circuit Diagram.

